



* April 5, 2001

+39 331 402 0145

✉ mattia.garatti

@universite-paris-saclay.fr

Mattia Garatti

Curriculum Vitae

I am a Ph.D. student in Mathematics interested in Mathematical Analysis, and its interaction with Mathematical Physics and Geometry. In my master thesis I have worked on the variational representations of optimal transport and Schrödinger problem: the aim has been to highlight how the first can be recovered as Γ -limit of the second. Currently, I am carrying out my research on regularized optimal transport problem(s).

Education

Ph.D. in Mathematics, *Université Paris-Saclay*, Orsay, France. **2025 – ongoing**

Thesis: "On regularized Optimal Transport problems". Supervisors: Luca Nenna - Simona Rota-Nodari.

Master degree in Mathematics (LM-40), *Università Cattolica del Sacro Cuore*, Brescia, 110 cum laude. **2023 – 2025**

Average: 30 cum laude. Main courses' topics: Real and Functional Analysis, Measure Theory, Differential Geometry, Mathematical Physics. Thesis: "Variational representations of optimal transport and Schrödinger problem". Supervisor: Luca Tamanini - Cosupervisor: Nicolò De Ponti.

Bachelor degree in Mathematics (L-35), *Università Cattolica del Sacro Cuore*, Brescia, 110 cum laude. **2020 – 2023**

Average: 31.33/30. Thesis: "Il Teorema di esistenza di Peano per le equazioni differenziali ordinarie", provision of a theory on ODEs in the Peano's assumptions. Supervisor: Marco Marzocchi

High School Diploma, *Liceo Scientifico 'Camillo Golgi'*, Breno (BS), **2015 – 2020**
100/100.

Talks

Variational representations of optimal transport and Schrödinger problem, **2025**
Math PhD Days at UniTO, Turin.

Talk on my master thesis work

Teaching

MathExcellence 2025: corso di eccellenza in Matematica, *Università Cattolica del Sacro Cuore*, Brescia (BS). **June 2025**

Assistance and tutoring activities (5 hours)

Teacher Assistant, internship at *Liceo Scientifico 'Golgi'*, Breno (BS). **September 2022**

Frontal lectures for high school students, exam administration and correction, laboratory assistant

Private Teacher, internship at *Cooperativa 'Assolo'*, Berzo Inferiore (BS). **August – September 2018**

Lessons of Mathematics and Physics for high school students with SLD

Teacher Assistant, *supervision of preschool children, scientific experiments*, **June 2018**
Fondazione Scuola Materna 'San Giuseppe', Gratacasolo (BS).

Work experience

Math writer, Zanichelli editore S.p.A., Bologna (BO).

2024 – 2025

Teamwork with the editorial staff, autonomous creation of exercises of various difficulty on different topics to be included in future publications

Summer schools and workshop attended

Geometry, duality and convexity in new OT problems, *IMO Paris–Saclay*, Orsay, France. **October 2025**

Workshop in new Optimal Transport problems arising in various applications

PGMO Days 2025, *EDF Lab Paris–Saclay*, Palaiseau, France. **October 2025**

Workshop on Optimal Trasport and its interactions with Data Sciences

Advances in Calculus of Variations and Nonlinear Analysis, *Università Cattolica del Sacro Cuore*, Brescia. **September 2025**

Workshop in honor of Marco Degiovanni's 70th birthday

Il Problema di Plateau, *Università Cattolica del Sacro Cuore*, Brescia. **April 2025**

Lecture series about different approach to Plateau problem(s)

Introduction to Mechanics of Swelling, *Università Cattolica del Sacro Cuore*, Brescia. **February 2025**

Lecture series about mathematical models for swelling

Introduzione ai modelli matematici per i cristalli liquidi, *Università Cattolica del Sacro Cuore*, Brescia. **February 2024**

Lecture series about mathematical models for liquid crystals

Interuniversity Mathematics Summer School, *University of Perugia*. **July – August 2023**

Complex Analysis and PDE courses with related final exams successfully passed

Mathematics in Ljubljana, *University of Ljubljana, FMF and IMFM*. **July 2023**

Summer School with courses on Formalization of Mathematics and Introduction to Complex Dynamics

Storia e tecniche dei metodi geometrici di rappresentazione: un'introduzione, *Università Cattolica del Sacro Cuore*, Brescia. **March 2023**

Mini-course about cartography and perspective

Un'Introduzione alla Teoria dei Grafi, *Università Cattolica del Sacro Cuore*, Brescia. **February 2023**

Mini-course about Graph Theory and its applications

Alcuni Aspetti della Matematica Antica, *Università Cattolica del Sacro Cuore*, Brescia. **March 2021**

Webinar about history of Mathematics and some greek geometry aspects

Corso di Eccellenza in Matematica, *Università Cattolica del Sacro Cuore*, Brescia. **June 2019**

Mini-course for high school students about different fields of Mathematics

Computer skills

Programming languages: Phyton, C, C++

Mark-up languages: LaTeX, HTML

Languages

Italian: Mother tongue

English: Professional proficiency

Awards

"Smart" study award , <i>Università Cattolica del Sacro Cuore</i> , Brescia (BS). Award for 100 best students of the university	2025
Merit scholarship , <i>Tenaris</i> , Dalmine (BG). Scholarships awarded on merit every year during high school and university	2017 – 2024
Merit scholarship , <i>Consorzio del Bacino Imbrifero Montano dell'Oglio</i> , Bergamo. Scholarships awarded on merit every year during university	2021 – 2024
Exemption from first year university fees , <i>Università Cattolica del Sacro Cuore</i> , Brescia. Exemption by merit	2020
Merit scholarship , <i>Comune di Costa Volpino</i> (BG). Scholarships awarded on merit for high school diploma	2020

Organization

Member of the student jury of Galileo Prize. Jury for award regarding scientific dissemination	2024
Member of Joint Teachers – Students Commission , <i>Università Cattolica del Sacro Cuore</i> , Brescia.	2024 – 2025
Staff member of Celebration of Mind, Notte dei Ricercatori, Venite a vedere la Terra che gira. Mathematical popularization days, <i>Università Cattolica del Sacro Cuore</i>	
Staff member of Disfida matematica. Mathematical team competition for high school students, local phase in Brescia	2023 – ongoing
Member of Ufficio di Sezione Elettorale , <i>Comune di Costa Volpino</i> (BG). Scrutineer and president of polling station	2021 – ongoing
Parish volunteer , <i>Parrocchia di "San Zenone"</i> , Gratacasolo. Organization of summer camps and member of the local choir with budget manager functions	2015 – ongoing

I hereby authorize the use of my personal data in accordance to the GDPR 679/16 - "European regulation on the protection of personal data".

November 24th, 2025

Mattia Garatti

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